



Production Information Systems

Chapter 6 : E-Business and Web-Enabled Databases

ISE Department
Year 4/ Level 7

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Outline

1. Introduction
2. An HTML tutorial : web page design example
3. HTML page Tags
4. HTML text body formatting tags
5. Alignment, positioning and font control
6. Linking to other web sites and web pages
7. Using images
8. Using cascading style sheets
9. Using Frames, Lists and Forms



1.Introduction

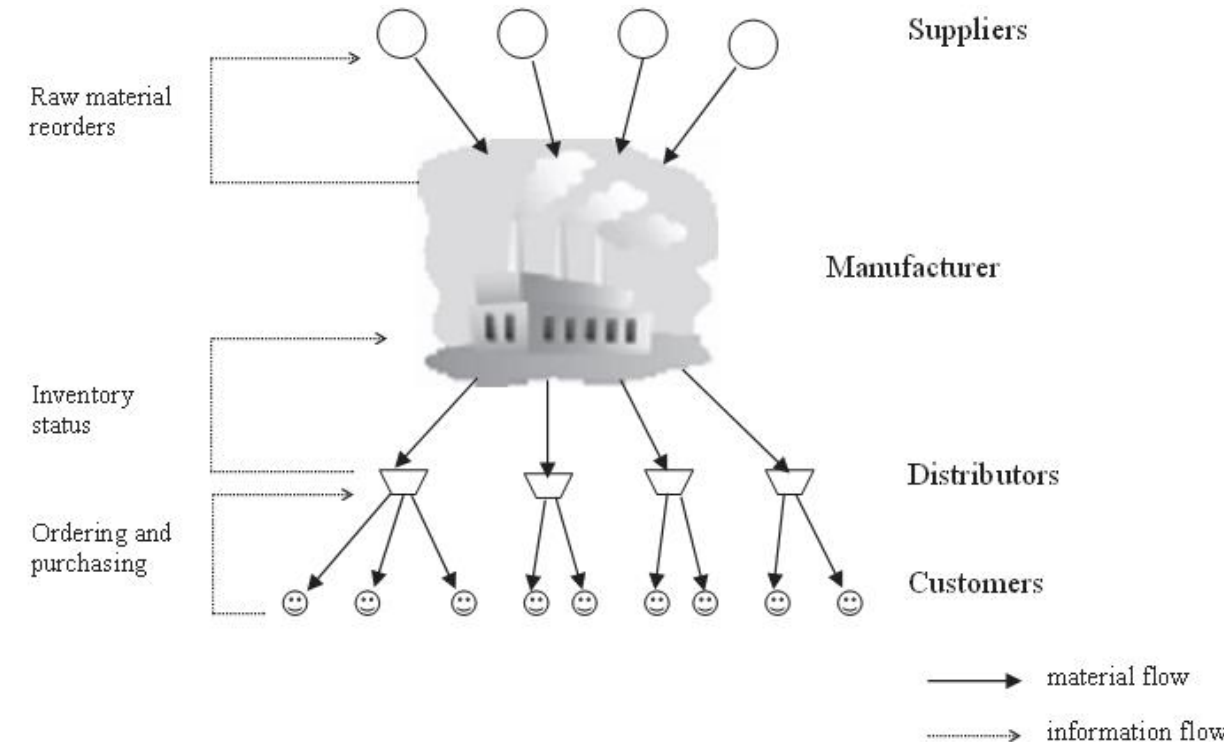
- At one time manufacturing managers scheduled and operated their plants using fairly long range planning horizons over which they forecasted demand.
- A typical approach was to isolate the plant operation from downstream activities (e.g., product distribution) by carrying buffer stocks of inventory in the plant warehouse and monitoring the level of those buffer stocks to decide when more inventory should be produced.
- All that changed with advances in information technologies. The developments in computer network technology and point-of-sale data acquisition technologies, such as bar code readers, made it possible to capture real-time information at final distribution points and to incorporate this information immediately into production planning and scheduling decisions at the plant.

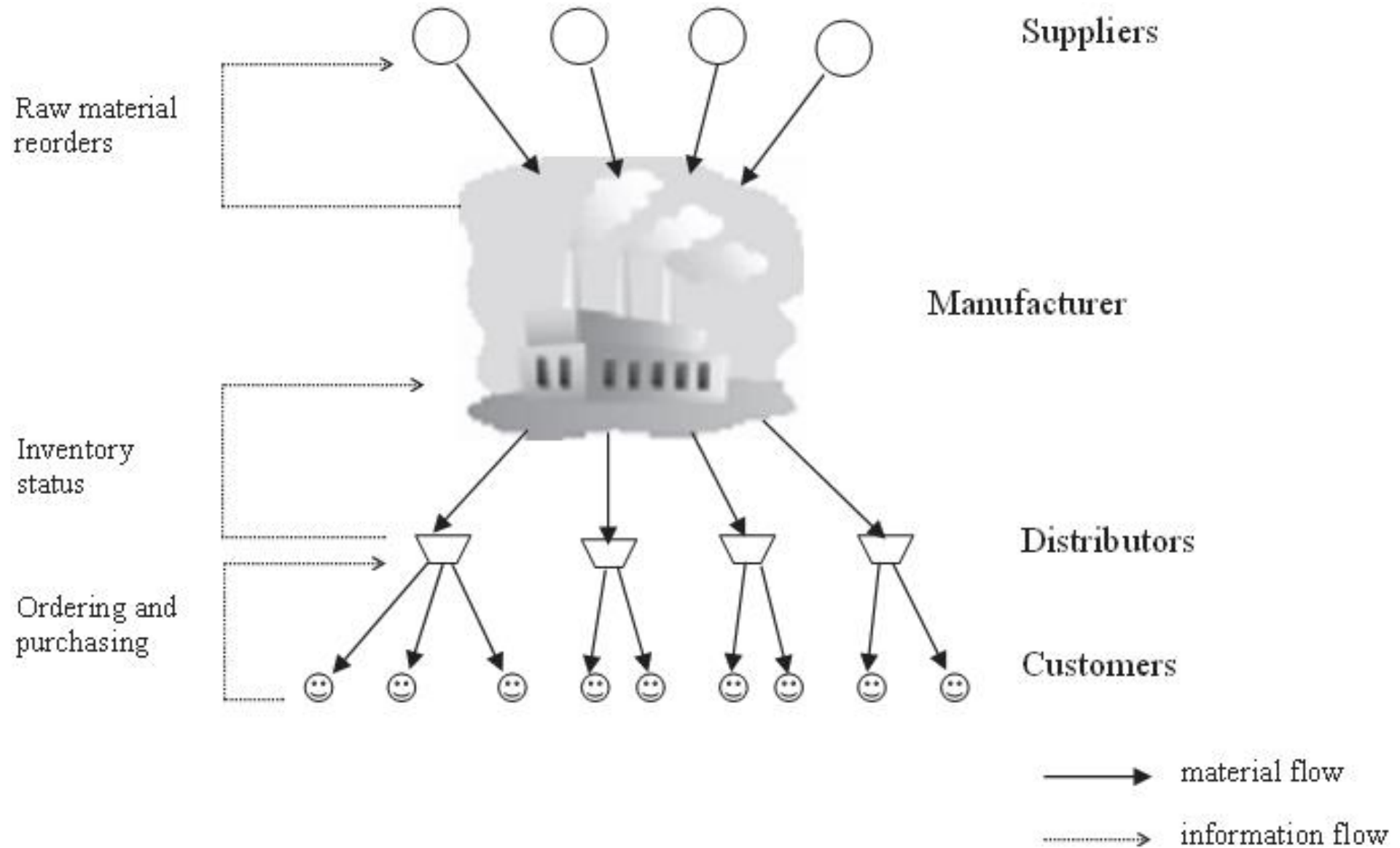
1.Introduction

- Thus, instead of monitoring inventory levels at the plant warehouse, it was possible to see the status of inventories throughout the distribution system and to anticipate when inventory stock replenishment orders would be placed.
- In some cases, information that exists beyond the enterprise can be made accessible to the plant for planning purposes.
 - For example, supermarkets have real-time inventory control through the use of bar code scanning at the checkout counters. The supplier (manufacturer) of food products may develop a relationship with the supermarket that allows the manufacturer to access the supermarket information system and extract information on current inventory levels.
- This allows better resupply forecasting and subsequent production scheduling decisions to be made by the manufacturer.

1.Introduction

- This vision of business relationships and information sharing has been incorporated into the concept of *supply chain management*.
- A ***supply chain*** is a set of related enterprises through which materials and associated information flow.
- A typical supply chain of a single manufacturer is shown.





1.Introduction

- It has an hourglass shape, with the manufacturing enterprise at the center.
- The lower part shows the physical distribution system from the manufacturing plant to the customer.
 - Here management is concerned with customer demand and appropriate inventory levels to service that demand at minimum cost.
 - Management problems in the distribution system also include transportation and the routing and scheduling of company-owned vehicles for delivery of the product to regional and local warehouses and customers.
- The upper level shows the input network of the supply chain.
 - Manufacturers are customers of suppliers, who themselves may be manufacturers with their own supply chains.
 - Problems on the input side include monitoring appropriate levels of raw materials and managing lead times between ordering and receiving shipments from suppliers so as not to disrupt production schedules due to delayed shipments

1.Introduction

- The supply chain is a useful context for the subject of e-business.
- Many of the problems of coordinating the operations of a supply chain arise from lack of timely information.
 - Today, companies are moving to a tightly integrated supply chain model in which information throughout the supply chain is immediately available to all participants.
 - The model popularized by Dell Computer is a case in point.
 - Customers order Dell's computer products directly over the Internet, and Dell uses this ordering information to procure parts over the Internet from its extensive Web of suppliers.
 - In this model, the connection between the plant floor and the broader supply chain is made possible by information exchange over the Internet.
- The term commonly used to describe this model is ***e-commerce***; here we use the associated term ***e-business***.

1.Introduction

- The purpose of information technology in e-business is to provide communication between the enterprise and its customers (**business to customer, or B to C**) and between the enterprise and its suppliers (**business to business, B to B**).
- The two main uses of e-business systems are
 - Intercompany transactions: includes customer order entry, purchasing from suppliers, and electronic billing, among others.
 - Information exchange: includes inventory level information sharing, quotation of lead times, and online checking on the status of an order, among others.
- Earlier electronic transactions between business partners (B to B) used a technology called ***electronic data interchange (EDI)***.
 - EDI enabled intercompany, computer-to-computer exchange of business documents in a standard format. With the advances in computer technology and the Internet in the late 1990s, business collaboration among enterprises shifted to Internet-based transaction processing.

1.Introduction

- Each enterprise in a supply chain partnership has its own enterprise resource planning (ERP) and manufacturing execution system (MES) and other software solutions that handle the internal business.
 - In the supply chain, these systems must work together with those of other supply chain partners to achieve the business objectives of the coalition of partners in the supply chain.
- From the perspective taken in this book, this raises an important issue.
 - A requirement for business partners to work together in a supply chain is that of database information exchange technologies and protocols.
 - Database information input and retrieval can be designed as part of an Internet application.
 - E.g., placing orders for books and other merchandise over the Internet.
 - Those applications are developed using a combination of hypertext markup language (HTML) for designing Web pages and some scripting language, such as VBScript or JavaScript, for interacting with the database.
 - This popular approach is used for placing orders in B-to-C applications

2. An HTML Tutorial

- There are also cases in which an enterprise may simply want to connect to a supplier via the Internet and transfer information from its database to that of the supply chain partner.
- The starting point for studying the details of Internet applications is to review HTML, which is the formatting language for Web pages.
- ***Hypertext markup language (HTML)*** is a “scripting” language, as opposed to a programming language. It is used to *mark up* the pages of electronic documents with formatting commands that can be easily interpreted by a Web browser.
- A Web browser is a computer program that interprets HTML commands according to an (evolving) HTML standard.

2. An HTML Tutorial

- Because HTML is basically a document page design tool, there are many commercial products that allow a user to directly design a Web page and automatically convert the design into HTML code so that it can be viewed on a browser.
 - Some leading products are FrontPage (by Microsoft) and Dreamweaver (by Macromedia).
- Because we are interested in implementing database queries on the Web, it would be appropriate for us to know some of the details of Web technology and HTML code.
 - Therefore, we will review a subset of HTML and see how it fits into the overall process of making our database forms Web enabled.

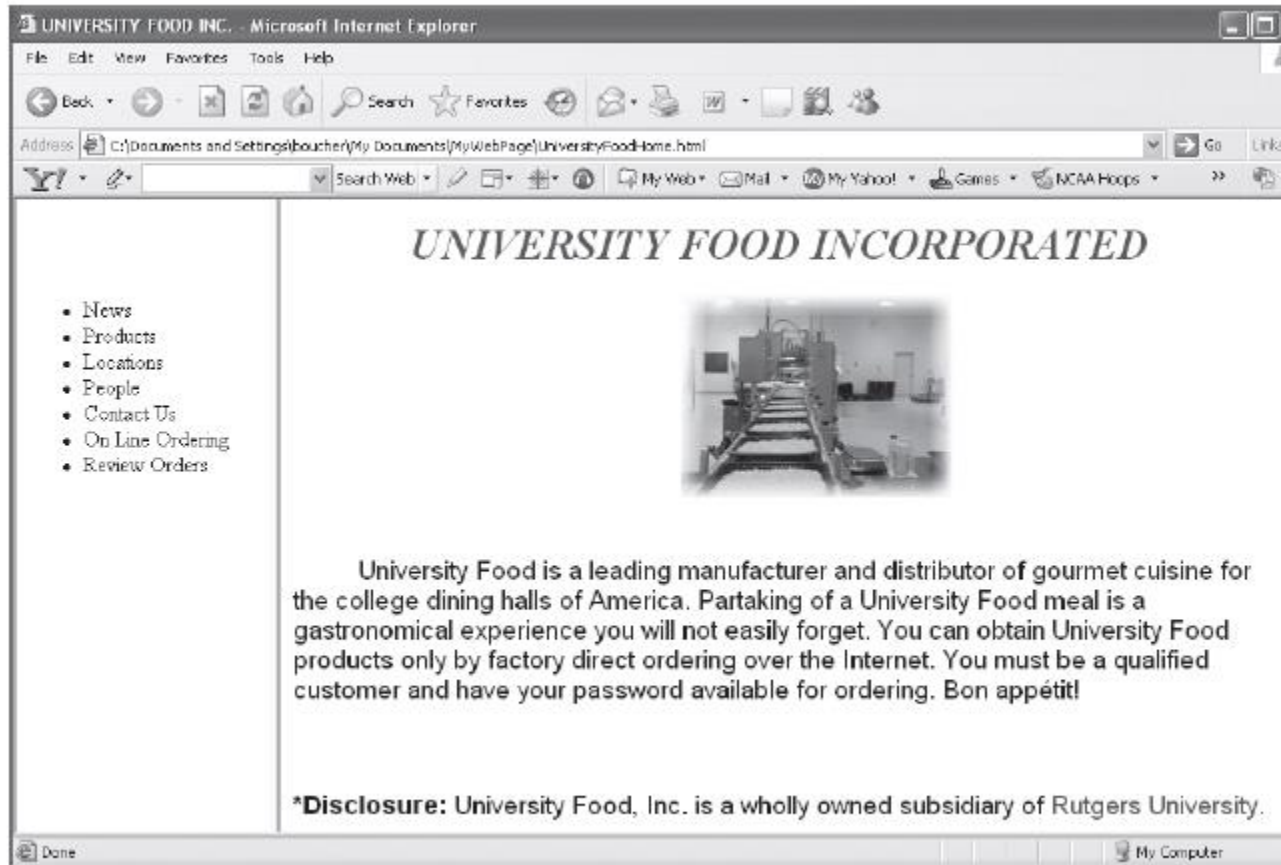
2. An HTML Tutorial

WEB PAGE DESIGN EXAMPLE

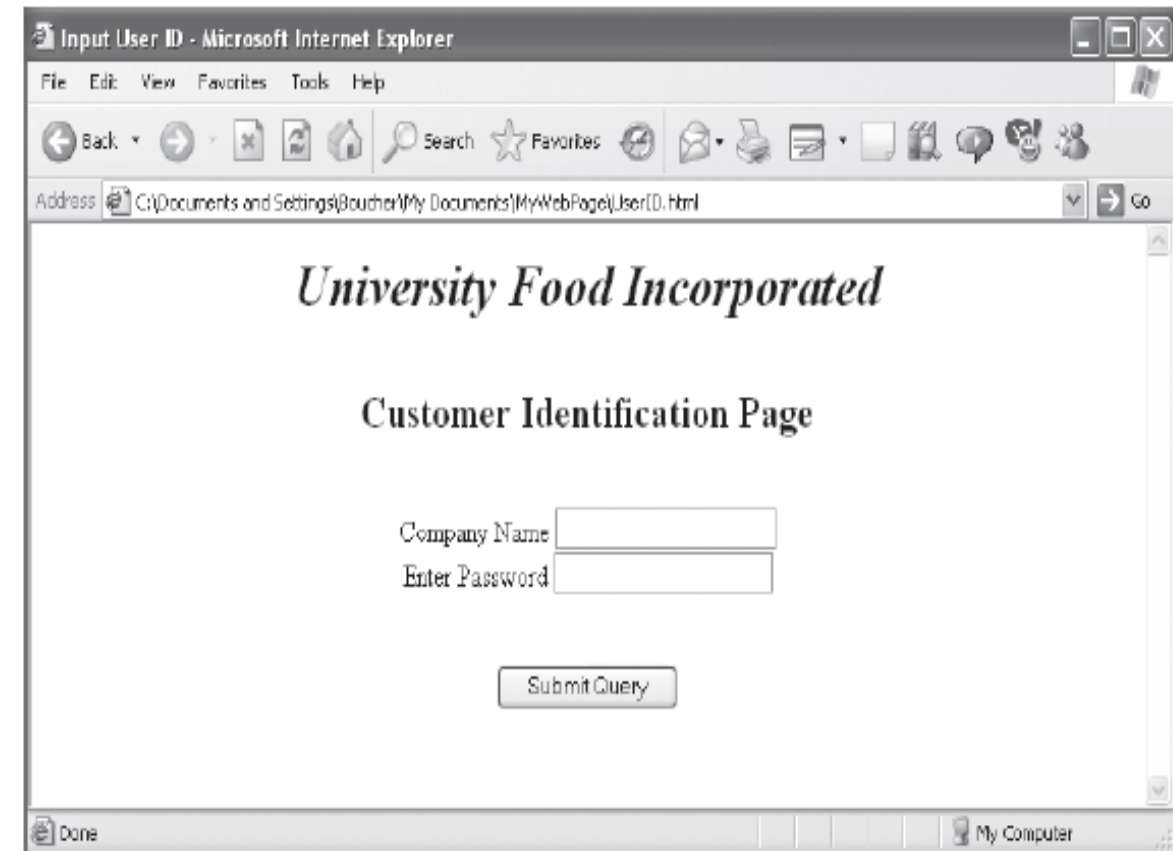
- We first must determine what the Web Page should look like and how it should function.
 - The set of Web pages that will be used for this illustration is based on a common function in B-to-C (business-to-customer) Internet transactions: entering an order for a product.
- Two screens will be involved:
 - An HTML Homepage with references to text and graphic files
 - A form for entering customer information

2. An HTML Tutorial

WEB PAGE DESIGN EXAMPLE



a) Homepage



b) Customer input form

Figure 8.2 Example Web pages.

There are two general classifications of Web page applications that are represented

These classes are “**static**” and “**dynamic**” Web pages, respectively.

2. An HTML Tutorial

WEB PAGE DESIGN EXAMPLE

- ***A static Web page*** consists of a file or a set of files that are programmed in HTML. The contents of these files are not expected to change very often, and they are not capable of changing their presentation on a Web browser based on a user query. The homepage is an example of a typical static Web page.
- The static Web page environment is a good example of **client-server architecture**.
 - The World Wide Web (WWW) consists of a network of computers that act in two roles. **The browser** on a workstation acts as a client.
 - Examples are Netscape Navigator and Windows Explorer.
 - **The server** is the computer and software that handles the client's requests for a particular Web page.
 - Typical example of server software for Web applications is Microsoft's Internet Information Server.

2. An HTML Tutorial

WEB PAGE DESIGN EXAMPLE

- ***Dynamic Web pages*** are more complicated. In that environment, a client can request information that may change based on the requestor or on the timing of the request. In other words, the information is dynamic.
 - A database search is an example of this kind of situation, as is the customer input form

3. An HTML Tutorial

HTML PAGE TAGS

- Remember: HTML is a language for simply marking up a document page.
- The marking up is done using HTML *tags*. A **tag** is a coded command used to define how part of a Web page is to be displayed on a browser.
- Tag syntax appears as words within angle brackets, as shown here: <word>. They delineate an area of the page for a specific kind of text.
 - For example, the title of the page is indicated by the start tag <TITLE>.
- A start tag usually requires an end tag to indicate the end of the assigned area.
 - The end tag begins with a foreward slash /.
 - Thus, the end of the area in which the title appears is indicated by </TITLE>.

3. An HTML Tutorial

HTML PAGE TAGS

- Some tags are used to give overall structure to a document. These structural format tags are as follows:
- `<HTML>` and `</HTML>`.
 - This tag set declares that the document within the tags is an HTML document. Normally, you will see these tags at the start and the end of the formatting instructions of the page.
- `<HEAD>` and `</HEAD>`.
 - The header tag defines the contents of the region of the page where general theme (header) information will appear.
- `<TITLE>` and `</TITLE>`.
 - The name of the Web page is placed within these tags. The title of the page is an example of header information. The title of the Web page will appear on the title bar of the Web browser when the page is opened.
- `<BODY>` and `</BODY>`.
 - This usually follows the header and is the area where text within the browser window will appear.

3. An HTML Tutorial

HTML PAGE TAGS

HTML documents can be created in any text editor. For the examples given in this chapter, we will use Notepad, the text editor that comes with the Windows operating system.

Example : Exercise HTML

HTML Exercise & 1:

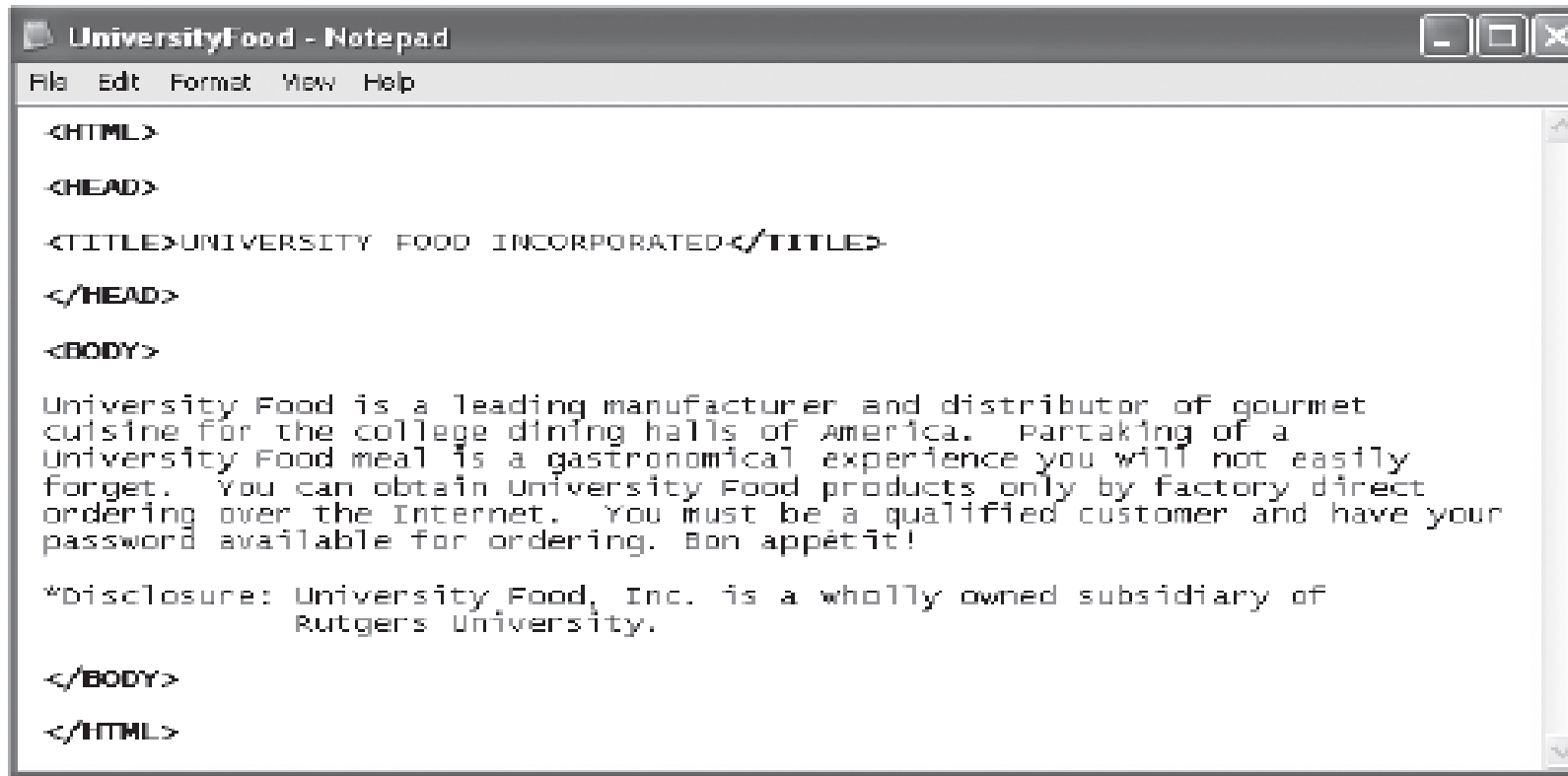
We wish to use the *structural formatting tags* of HTML to illustrate a default page based on Figure 8.2a.

1. Create a folder that you will use to save your web pages (for example, C:/MyWebPage).
2. Open the Notepad text editor (*Accessories => Notepad*).
3. Type the text shown in Figure 8.4.
4. Save the file as type "*All files*" with the File name *UniversityFood.html* in your folder.

3. An HTML Tutorial

HTML PAGE TAGS

Example : Exercise HTML



```
UniversityFood - Notepad
File Edit Format View Help

<HTML>

<HEAD>

<TITLE>UNIVERSITY FOOD INCORPORATED</TITLE>

</HEAD>

<BODY>

University Food is a leading manufacturer and distributor of gourmet
cuisine for the college dining halls of America. Partaking of a
University Food meal is a gastronomical experience you will not easily
forget. You can obtain University Food products only by factory direct
ordering over the Internet. You must be a qualified customer and have your
password available for ordering. Bon appétit!

*Disclosure: University Food, Inc. is a wholly owned subsidiary of
Rutgers University.

</BODY>

</HTML>
```

Figure 8.4 Using structural formatting tags.

3. An HTML Tutorial

HTML PAGE TAGS

Example : Exercise HTML

- Note that it is important to save the file as type “all files.” If you save it as a text file, the browser will not recognize it as an HTML file.
- To test the functionality of your page, open a Web browser (Netscape Navigator or Internet Explorer). Browse to the folder and open the file in your browser.
- Note that the title of the page appears in the title bar of the browser. It does not show up on the Web page. All the text that the viewer will see in the browser window must be placed between the <BODY> </BODY> tags.
- Also, the text is not formatted; it runs on continuously, regardless of how it was typed. To control the detailed formatting of the text, it is necessary to include formatting tags.

3. An HTML Tutorial

HTML PAGE TAGS

- *Example : Exercise HTML*
- Here are some of the most basic formatting tags for the text appearance on the page:
 - `<Hx>` and `</Hx>`.
 - Headings that appear in the Web browser window (placed between `<BODY>` and `</BODY>` tags) are typically larger than the other text in the body area. The easiest way to create a heading larger than the default body text size is to use the heading tags: `<H1>` `</H1>`, or `<H2>` `</H2>`, and so forth. The index indicates the relative size of the heading text, where H1 is the largest and H2 through H6 become progressively smaller.
 - `<P>` and `</P>`.
 - These tags enclose a paragraph. Within the paragraph, the text is continuous when opened in the browser window. The closing tag, `</P>` is optional.
 - `<BR>`.
 - This is a line break. It is placed at the end of a section where a line break is desired. It is a single tag and does not require a closing tag.

4. An HTML Tutorial

HTML TEXT BODY FORMATTING TAGS

Example : Exercise HTML

To illustrate the use of these tags and to render a better appearance to the Web page, open UniversityFood.html and edit the text in accordance with the instructions of HTML Exercise 8.2. The new HTML tags are shown in bold in Figure 8.6.

HTML Exercise 8.2:

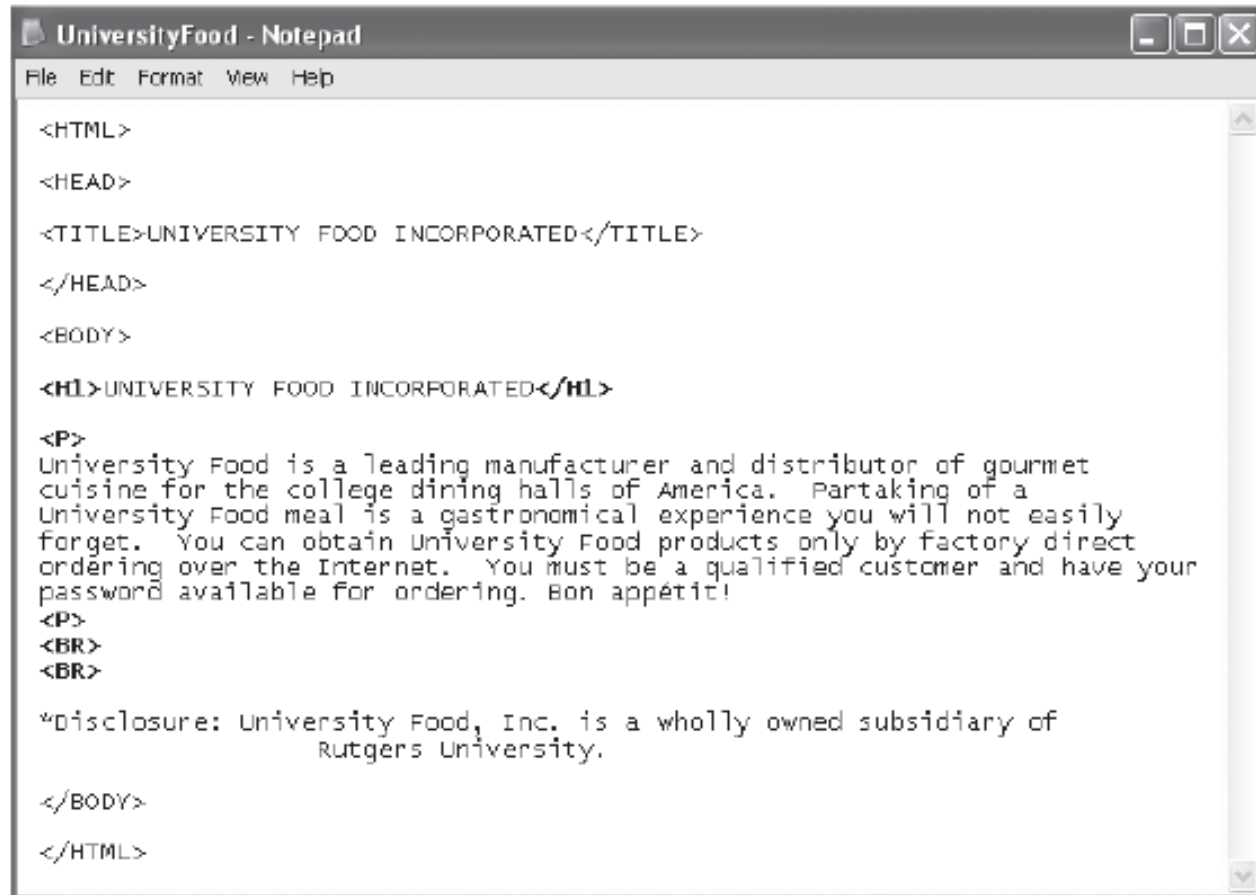
We wish to add more structure to Figure 8.5 using ***basic formatting tags***.

1. Open the file ***UniversityFood.html*** in Notepad.
2. Add heading tags, a paragraph tag, and line breaks as shown in Figure 8.6. Save and close the file.
3. Using Netscape Navigator or Internet Explorer, browse to the file and open it in the browser window. It should appear like Figure 8.7.

4. An HTML Tutorial

HTML TEXT BODY FORMATTING TAGS

Example : Exercise HTML



```
<HTML>
<HEAD>
<TITLE>UNIVERSITY FOOD INCORPORATED</TITLE>
</HEAD>
<BODY>
<H1>UNIVERSITY FOOD INCORPORATED</H1>
<P>
University Food is a leading manufacturer and distributor of gourmet
cuisine for the college dining halls of America. Partaking of a
University Food meal is a gastronomical experience you will not easily
forget. You can obtain University Food products only by factory direct
ordering over the Internet. You must be a qualified customer and have your
password available for ordering. Bon appétit!
<P>
<BR>
<BR>
*Disclosure: University Food, Inc. is a wholly owned subsidiary of
Rutgers University.
</BODY>
</HTML>
```

Figure 8.6 Adding text formatting tags.

Note the improved control over the layout of the page. Also, the page caption now appears as a heading of the desired size in the browser window. Heading tags within the body of the page result in a caption on the page within the browser window.

5. An HTML Tutorial

ALIGNMENT, POSITIONING, AND FONT CONTROL

- Despite the improvement in the format of the Web page, there are still some obvious deficiencies in the appearance, alignment, and positioning of the text.
- There are some additions to text formatting tags. These additions are called *attributes*.
 - An **attribute** is a command, placed within the format tags, to further define a feature of the text within these tags.
 - and . These tags indicate that the text appearing between them should be bold.
 - <I> and </I>. These tags indicate that the text appearing between them should be in italic.
 - <P Align="Right">, <P Align="Center">, <P Align="Left">. The command *Align* is an attribute of the P tag or the Hx tag. It specifies the alignment of the paragraph or heading. If alignment is not specified, the text defaults to <P Align="Left">.

5. An HTML Tutorial

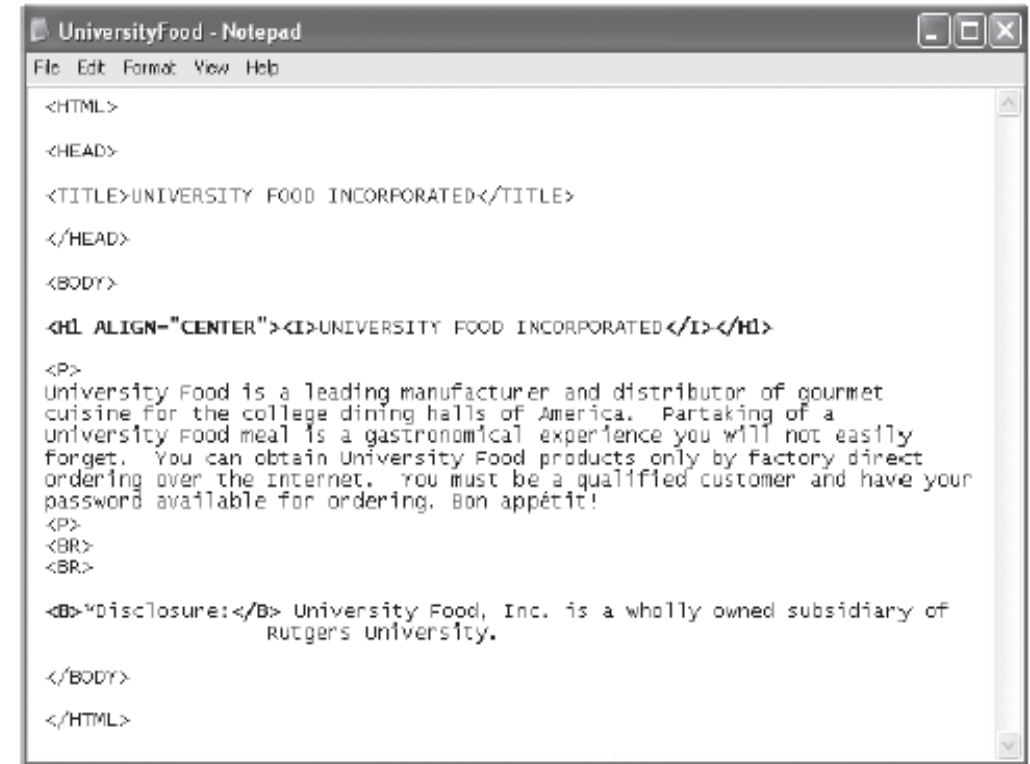
ALIGNMENT, POSITIONING, AND FONT CONTROL

To illustrate these tags, open the UniversityFood.html file and add the attributes as shown in Figure 8.8.

HTML Exercise 8.3:

We wish to add more definition to Figure 8.7 by using *tag attributes*.

1. Open the file *UniversityFood.html* in Notepad.
2. Add the tags for bold, italics, and align as shown in Figure 8.8. Save and close the file.
3. Using Netscape Navigator or Internet Explorer, browse to the file and open it in the browser window. It should appear like Figure 8.9.



```
<HTML>
<HEAD>
<TITLE>UNIVERSITY FOOD INCORPORATED</TITLE>
</HEAD>
<BODY>
<H1 ALIGN="CENTER"><I>UNIVERSITY FOOD INCORPORATED</I></H1>
<P>
University Food is a leading manufacturer and distributor of gourmet
cuisine for the college dining halls of America. Partaking of a
University Food meal is a gastronomical experience you will not easily
forget. You can obtain University Food products only by factory direct
ordering over the internet. You must be a qualified customer and have your
password available for ordering. Bon appétit!
<P>
<BR>
<BR>
<B>*Disclosure:</B> University Food, Inc. is a wholly owned subsidiary of
Rutgers University.
</BODY>
</HTML>
```

Figure 8.8 Adding alignment, positioning, and font attributes.

6. An HTML Tutorial

LINKING TO OTHER WEB SITES AND WEB PAGES

- The power of ***hypertext*** is the ability to link pages within a document and to link to other Web pages on other machines on the World Wide Web.
- A hyperlink is a command that, when executed, opens another file, browser window, or Web page. Linking to other pages and other documents on the Internet is easy using the ***anchor*** `<A>` tag.
- The anchor tag uses a format that begins with `<A HREF=` and is followed by the page file location or the Web address of the page to which the link is directed.
- For example, if one is linking to a file named “home.html” that resides within the same folder as the current page, the anchor would be as follows:

`<A HREF=“home.html”>`

6. An HTML Tutorial

LINKING TO OTHER WEB SITES AND WEB PAGES

- If the linked page (file) is in a separate folder, the complete path to the file must be given within quotations. If the link is to another Web site on the WWW, it can be referred to by its WWW address. For example, in the case of Rutgers University, `<A HREF="http://www.rutgers.edu">`.
- When you create a link, you do not have to display the target URL address in the browser window, although you can if you want to. For example, we have a link to the Rutgers Web site that just reads "Rutgers University." This can be done by adding text after the opening anchor and adding a closing anchor as follows:

... subsidiary of `<A HREF="http://www.rutgers.edu">` Rutgers University. `</A>`

- In this format, the words "Rutgers University" are displayed as a hyperlink. Clicking on the hyperlink in the browser window with a mouse will display `http://www.rutgers.edu` in the Web browser URL panel, and the new Web page will be loaded.

7. An HTML Tutorial USING IMAGES

- Images are popular on Web pages to gain viewer interest. Images must be created separately and stored as either .jpg or .gif files.
- HTML simply formats the page and indicates the placement of the image on the page. When a user opens the Web page, the image file is called and it opens in the formatted location. The following tag is used with images.
- `<IMG>`.
 - This tag alerts the browser that an image should be located in the position on the screen where the tag occurs. Reference is made to the image file (.jpg or .gif) by pointing to the source file as follows:

`<IMG SRC="ImageName.gif">`

7. An HTML Tutorial

USING IMAGES

- If the image file is not located in the same folder as the file of the Web page calling it, the IMG tag must include the complete path to the file within the quotations.
- If the image is stored in a separate folder, the path to the file must be placed within the quotations.
- Image size can be controlled by using the dimension tags to scale the image.
 - For example, if the original image is 300 × 200 pixels, its size can be scaled to half using the HEIGHT and WIDTH specification tags as follows:

7. An HTML Tutorial USING IMAGES

HTML Exercise 8.4:

We wish to add the *image link* and *anchor tag* to Figure 8.9.

1. Open the file *UniversityFood.html* in Notepad.
2. Add the image link and anchor tag as shown in Figure 8.10. Save and close the file.
3. Using Netscape Navigator or Internet Explorer, browse to the file and open it in the browser window. It should appear like Figure 8.11.



```
<HTML>
<HEAD>
<TITLE>UNIVERSITY FOOD INCORPORATED</TITLE>
</HEAD>
<BODY>
<H1 ALIGN="CENTER"><I>UNIVERSITY FOOD INCORPORATED</I></H1>
<P ALIGN="CENTER">
<IMG SRC="foodpic.jpg">
</P>
<BR>
<P>
University Food is a leading manufacturer and distributor of gourmet
cuisine for the college dining halls of America. Partaking of a
University Food meal is a gastronomical experience you will not easily
forget. You can obtain University Food products only by factory direct
ordering over the internet. You must be a qualified customer and have your
password available for ordering. Bon appetit!
</P>
<BR>
<BR>
<B>*disclosure:</B> university Food, Inc. is a wholly owned subsidiary of
<A HREF="http://www.rutgers.edu">Rutgers University.</A>
</BODY>
</HTML>
```

Figure 8.10 Adding an anchor and an image.

8. An HTML Tutorial

USING CASCADING STYLE SHEETS

- More control can be exercised over the appearance of a document by using **style sheets**.
- Style sheets have been used for many years in desktop publishing to control elements of appearance, such as font size, color, and indentation.
- A style sheet uses methods that are specified as arguments of a tag.
- The Web browser reads these arguments just as it reads any other HTML tag instruction and displays the text according to the desired style.
- There are three ways of using style sheets:
 - Inline
 - Embedded
 - External

8. An HTML Tutorial

USING CASCADING STYLE SHEETS

- The *inline* approach simply adds the argument to the tag that appears in the body of the HTML file.
- If it is desirable to have the text beneath the image to appear indented and the text font to be 14 point Arial, the opening paragraph tag would be augmented as follows:

<P STYLE="text-indent: 0.75in;font: 14pt arial">

- The keyword STYLE indicates that the instructions that follow are style methods.
 - In this case, the text-indent method and the font method are used.
- When the browser encounters the STYLE keyword, it executes the methods that follow on the relevant text.

8. An HTML Tutorial

USING CASCADING STYLE SHEETS

- Inline STYLE can also be useful for aligning text on a page.
- A special case is its use in conjunction with the <ADDRESS> tag that provides a single spaced address block.

- For example,

```
<ADDRESS STYLE="text-indent: 2.00in">NAME</STYLE></ADDRESS>  
<ADDRESS STYLE="text-indent: 2.00in">STREET</STYLE></ADDRESS>, etc.
```

- This will provide a left-aligned, single-spaced, indented address block.
- The inline STYLE statement can be used with other HTML tags. It can be used with:
 - Block formatting tags (<BODY>, <CENTER>, etc.)
 - List tags (,)
 - Image ()
 - Anchor (<A>) tags

8. An HTML Tutorial

USING CASCADING STYLE SHEETS

- Using the ***embedded*** style sheet format, the user places all style information in a style declarative section before the body of the text.
- The style declarative section is bounded by a <STYLE> </STYLE> tag set and is placed just after the header information.
- For example, in Figure 8.10, an embedded style sheet section could be added as follows:

```
<HTML>
<HEAD>
<TITLE>UNIVERSITY FOODS INCORPORATED</TITLE>
</HEAD>
<STYLE>
BODY {margin-left: 0.5in;margin-right: 0.5in}
      H1 {color: blue}
      P {text-indent: 0.5in;font: 14pt arial}
      A {text-decoration: none}
</STYLE>
```

8. An HTML Tutorial

USING CASCADING STYLE SHEETS

- The preceding style sheet sets left and right margins for the entire body of the document.
 - Each time the `<H1> </H1>` tag set is encountered, the header text will be colored blue.
 - Each time a `<P> </P>` tag set is encountered, a 0.5-inch indentation will appear on the first line of the paragraph and the font will be 14 point Arial.
 - Finally, text enclosed in anchor tags, `<A> </A>`, will appear without any special decoration. That is to say, no italics, bold, or underline will appear.
- By default, anchor tags appear with underlines and are colored blue. To alter the color, a color method can be added to the style sheet anchor tag argument.
- Finally, it is possible to define the style sheet as a separate document, called an ***external*** style sheet. Under this format, each Web page will consist of two files. The first file is the .html file that defines the contents of the Web page. A second file is created that contains the style sheet.

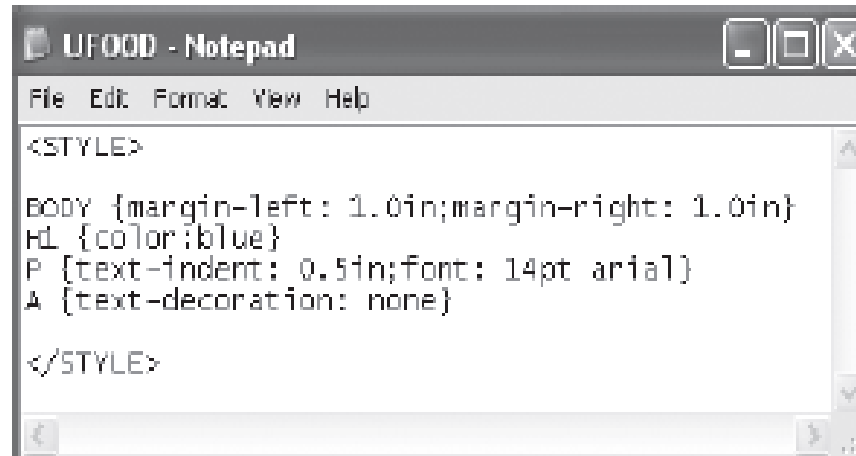
8. An HTML Tutorial

USING CASCADING STYLE SHEETS

- This second file is saved as a .css (cascading style sheet) file. The .css file is referenced from the .html file.
- When the Web browser encounters the reference while displaying the Web page, it opens the .css file and incorporates the style sheet methods contained therein into the appearance of the Web page.
- an example of an external style sheet named UFOOD.css. To link to the style sheet in Figure 8.12, the Web page (.html file) must declare the linking information in the header section of the file. This is shown in Figure 8.13 for the University Food Incorporated home page.

8. An HTML Tutorial

USING CASCADING STYLE SHEETS



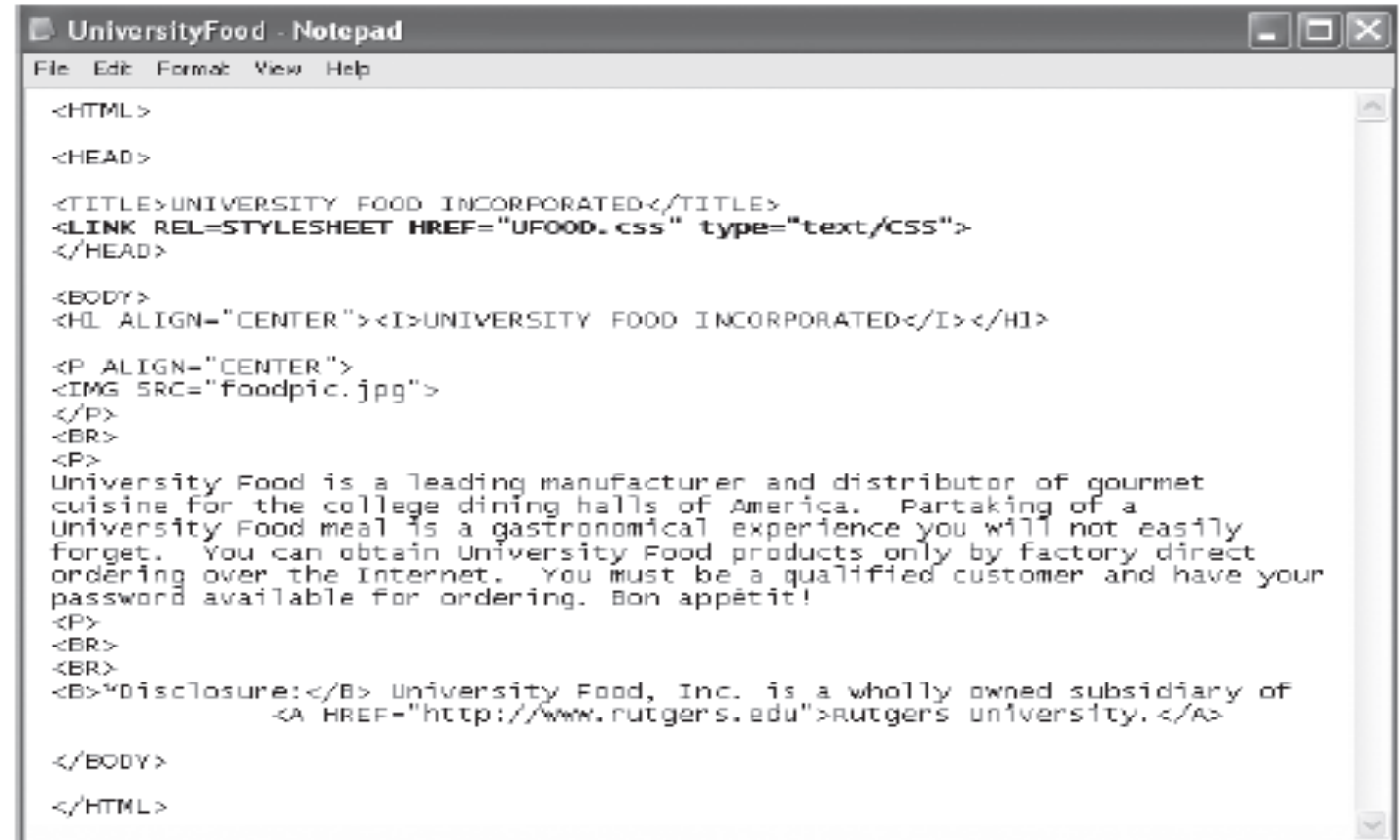
```
<STYLE>

BODY {margin-left: 1.0in;margin-right: 1.0in}
H1 {color:blue}
P {text-indent: 0.5in;font: 14pt arial}
A {text-decoration: none}

</STYLE>
```

Figure 8.12 External style sheet.

- An example of an external style sheet named UFOOD.css. To link to the style sheet in the Web page (.html file) must declare the linking information in the header section of the file. This is shown in Figure 8.13 for the University Food Incorporated home page.



```
<HTML>

<HEAD>

<TITLE>UNIVERSITY FOOD INCORPORATED</TITLE>
<LINK REL=STYLESHEET HREF="UFOOD.css" type="text/CSS">
</HEAD>

<BODY>
<H1 ALIGN="CENTER"><I>UNIVERSITY FOOD INCORPORATED</I></H1>

<P ALIGN="CENTER">
<IMG SRC="foodpic.jpg">
</P>
<BR>
<P>
University Food is a leading manufacturer and distributor of gourmet
cuisine for the college dining halls of America. Partaking of a
University Food meal is a gastronomical experience you will not easily
forget. You can obtain University Food products only by factory direct
ordering over the Internet. You must be a qualified customer and have your
password available for ordering. Bon appétit!
<P>
<BR>
<BR>
<B>"Disclosure:</B> University Food, Inc. is a wholly owned subsidiary of
<A HREF="http://www.rutgers.edu">Rutgers university.</A>

</BODY>

</HTML>
```

Figure 8.13 Web page with link to style sheet.

8. An HTML Tutorial

USING CASCADING STYLE SHEETS

- The <LINK> tag opens the link. The arguments declare it is a link to a style sheet, with the reference file name “UFOOD.css,” and is a text type file with a .css extension.
- Note that the HREF argument of the LINK tag is just the style sheet file name. This indicates that it is in the same folder as the file that calls it.
- If it is installed in a different location, the complete path to the style sheet file must be included within the quotations.
- The external style sheet format allows the user to separate style settings from the main Web page document.
- As a result, the same style sheet can be used over and over again for different Web pages and changes to the style sheet can be made without making changes in each Web page file.

8. An HTML Tutorial

USING CASCADING STYLE SHEETS

HTML Exercise 8.5:

We wish to create a *cascading style sheet* and reference it from `UniversityFood.html`.

1. Open Notepad and prepare the file `UFOOD.css` as shown in Figure 8.12. Save the file in your folder (MyWebPage) under Type “All files” as `UFOOD.css`.
2. Open the file *UniversityFood.html* in Notepad.
3. Add the link to the style sheet as shown in Figure 8.13. Save and close the file.
4. Using Netscape Navigator or Internet Explorer, browse to the file and open it in the browser window. Note the incorporation of the style sheet appearance.

9. An HTML Tutorial USING FRAMES

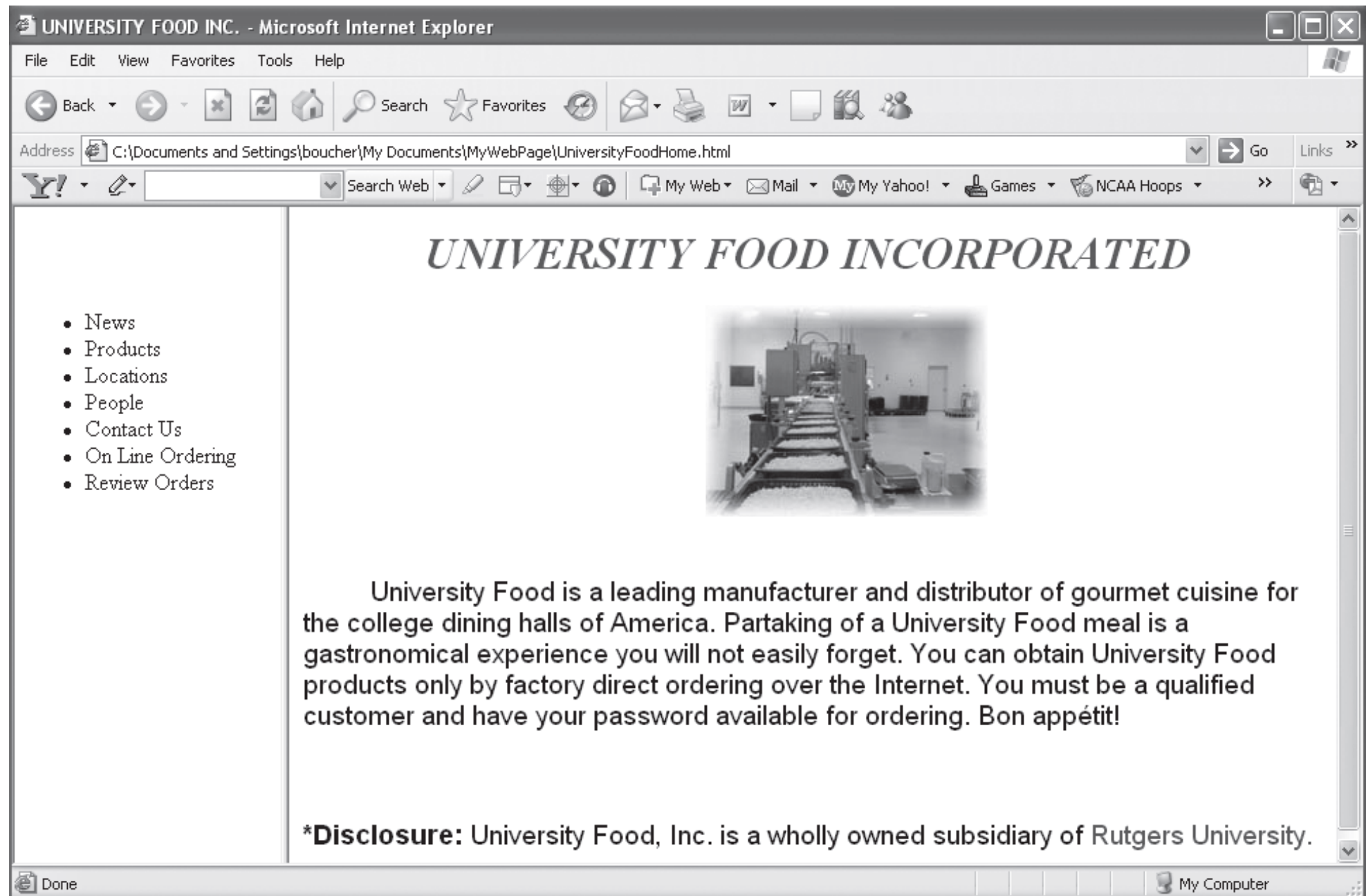
- Framed layouts are used to divide the browser window into multiple regions, known as *frames*.
- Only a few tags are required to implement frames:
 - <FRAMESET>. This tag is used to divide the browser window into frames.
 - There are two attributes that can be used with frameset; they are *rows* and *cols*.
 - Rows: divides the browser screen into multiple rows
 - Columns: divides the screen into multiple columns.
 - A typical syntax is as follows:

<FRAMESET Cols="20%, 80%">

...

</FRAMESET>

Dr. Jomana Bashatah



9. An HTML Tutorial USING FRAMES

- This command tells the browser to divide the browser window into two columns. The first column, starting from the left, comprises 20% of the screen, and the second column comprises 80% of the screen.
- The <FRAMESET> tag simply divides the browser window into multiple windows. The content of each window is separately determined by relating the window to the file that will give it content.
- To relate the frames to their contents, you need the <FRAME> tag:
 - <FRAME>. The *frame* tag tells the Web browser where to locate the contents of the frame.

9. An HTML Tutorial USING FRAMES

- Typical syntax is as follows:

```
<FRAMESET Cols="20%, 80%"> <!--make 2 column frames-->
```

```
<!--Assign content to frame #1-->
```

```
<FRAME SRC="List.html" FRAMEBORDER=1>
```

```
<!--Assign content to frame #2-->
```

```
<FRAME SRC="UniversityFood.html" FRAMEBORDER=1>
```

```
</FRAMESET>
```

- The FRAMEBORDER attribute is used to determine whether or not there will be a border around the frame. The value "1" indicates a border, and "0" indicates no border.

9. An HTML Tutorial USING FRAMES

- The preceding code includes some “comments.” To document the purpose of an HTML page, it may be appropriate to add comments to the page. Comments on an HTML page are not displayed in the browser window. They are strictly there for documentation purposes.
- Comments are added as follows:

`<!--This is a comment -->`

- To illustrate the use of the frameset and frame tags, two new files will be created: University-FoodHome.html and List.html.
 - The first file will become the University Food home page and will be partitioned into framesets.
 - The second file will become the left side panel.

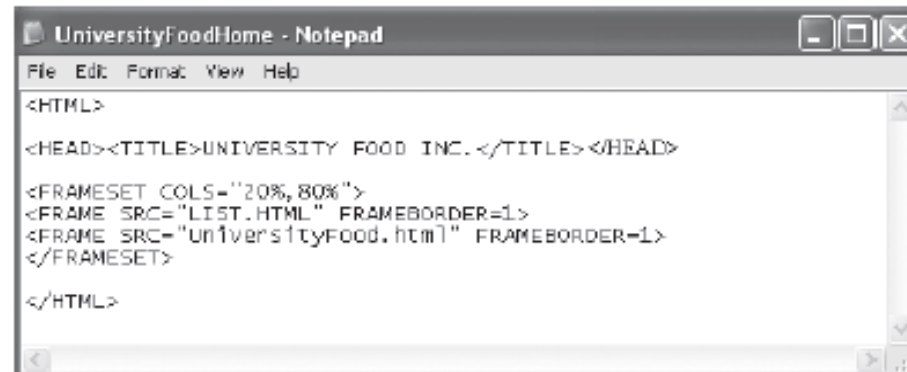
9. An HTML Tutorial USING FRAMES

Example :

HTML Exercise 8.6:

We wish to create a container Web page with frames for holding the contents of *UniversityFood.html*.

1. Open Notepad and prepare the file for *UniversityFoodHome.html* as shown in Figure 8.14. Save the file in your folder (MyWebPage) under Type "All files" as *UniversityFoodHome.html*.
2. Create a new file in Notepad. Leave the file blank. Save the file in your folder (MyWebPage) under Type "All files" as *List.html*.
3. Using Netscape Navigator or Internet Explorer, browse to the file *UniversityFoodHome.html* and open it in the browser window. It should look like Figure 8.15.



```
File Edit Format View Help
<HTML>
<HEAD><TITLE>UNIVERSITY FOOD INC.</TITLE></HEAD>
<FRAMESET COLS="20%, 80%">
<FRAME SRC="LIST.HTML" FRAMEBORDER=1>
<FRAME SRC="UniversityFood.html" FRAMEBORDER=1>
</FRAMESET>
</HTML>
```

Figure 8.14 University food home page with column frames.

9. An HTML Tutorial USING FRAMES

- When framesets are used with anchor tags, it is necessary to direct the Web browser to the frame in which to load new content from the hyperlink.
 - For example, List.html will eventually contain candidates for hyperlinks to other Web pages. In a typical implementation, the hyperlink will change content in the right-hand side frame, leaving the list of selection options on the left side unchanged.
- When a frameset is established in the container Web page, and there are hyperlinks for changing content, it is necessary to specify a name for the frame that will receive the content. If this is not done, the new content will be loaded into the frame on which the anchor tag resides.

9. An HTML Tutorial USING FRAMES

- HTML provides some keywords to control the loading of new content in a frameset.
- The keyword “NAME” is used to provide an identity for a frame.
- The name refers to the frame and not to the contents of the frame. So, for example, were we to name the right side frame of the frameset in UniversityFoodHome.html as “Rside,” the appropriate code would be as follows:

```
<FRAMESET Cols="20%, 80%">  
    <FRAME SRC="List.html" NAME="Lside" FRAMEBORDER=1>  
    <FRAME SRC="UniversityFood.html" NAME="Rside" FRAMEBORDER=1>  
</FRAMESET>
```

9. An HTML Tutorial USING FRAMES

- In future, the right side frame can be addressed by the name “Rside.” Then, for each anchor tag, the target frame for loading new content is identified by the keyword “TARGET.”
- For example, the hyperlink for products could be a file that lists the various products sold by the company.
- The anchor tag would indicate the target frame for loading the page, for example, as follows:
Products
- When the Web browser encounters the keyword TARGET, it knows that the argument is the name of the frame in which to load the content.

9. An HTML Tutorial USING FRAMES

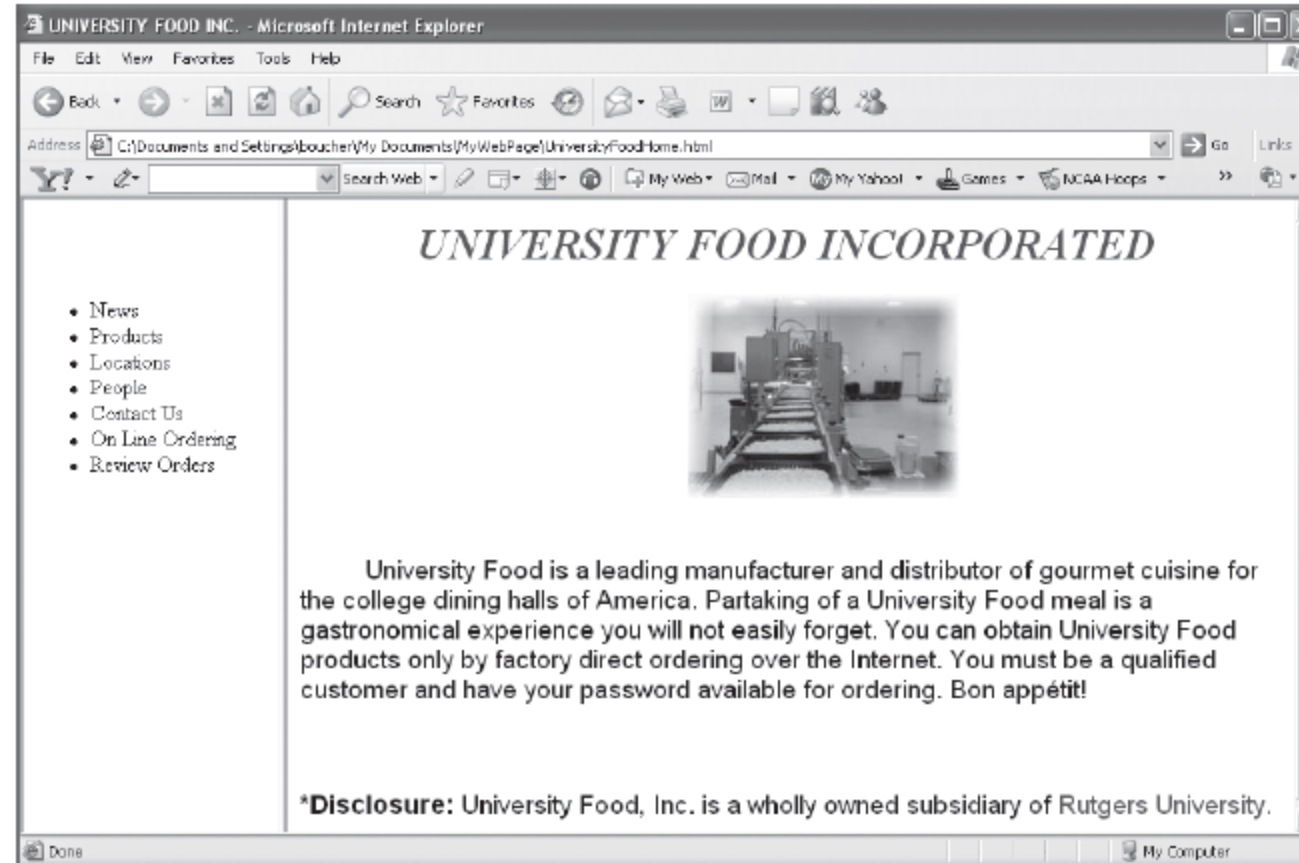


Figure 8.17 Final home Web page.

9. An HTML Tutorial

LISTS

- The left panel of shows a list of user options.
- Each option will have its own anchor tag that is a hyperlink to another Web page. Here we will simply set up the list.
- HTML includes tags that allow the user to set up numbered (ordered) and bulleted (unordered) lists. The tags are as follows:
 - `<OL>` and `</OL>`. These tags indicate the beginning and the end of an ordered list.
 - `<UL>` and `</UL>`. These tags indicate the beginning and end of an unordered list.
 - `<LI>`. This tag is placed in front of each item to be either numbered (ordered list) or bulleted (unordered list).

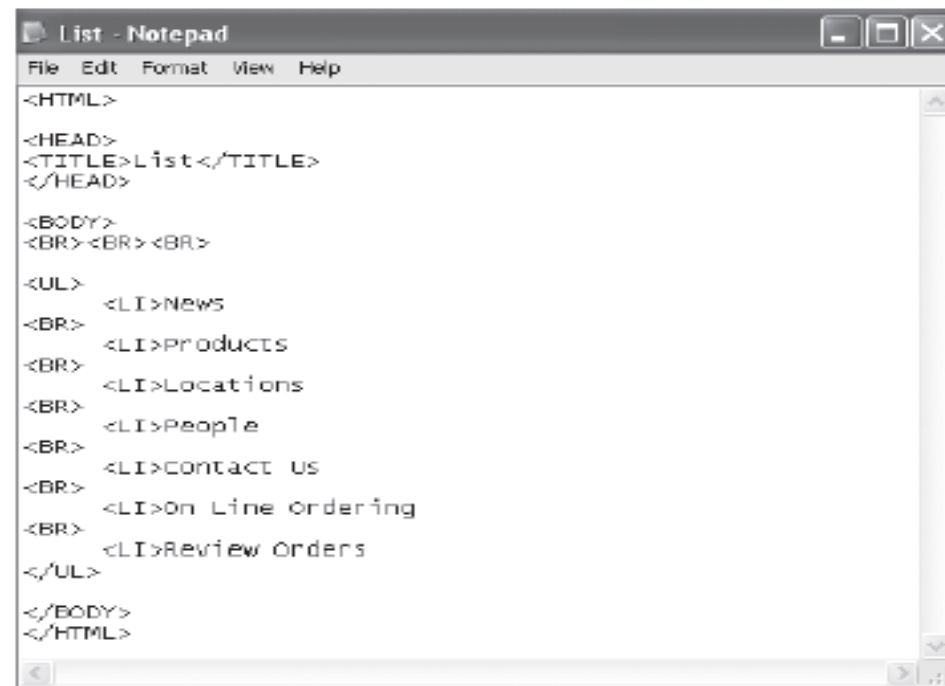
9. An HTML Tutorial

LISTS

HTML Exercise & 7:

We wish to create the left frame of Figure 8.15.

1. Open the file **List.html** in Notepad.
2. Create the file contents as shown in Figure 8.16. Save and close the file.
3. Using Netscape Navigator or Internet Explorer, browse to the file **UniversityFoodHome.html** and open it in the browser window. It should look like Figure 8.17.



```
File Edit Format View Help
<HTML>
<HEAD>
<TITLE>List</TITLE>
</HEAD>
<BODY>
<BR><BR><BR>
<UL>
<LI>News
<BR>
<LI>Products
<BR>
<LI>Locations
<BR>
<LI>People
<BR>
<LI>Contact Us
<BR>
<LI>On Line Ordering
<BR>
<LI>Review Orders
</UL>
</BODY>
</HTML>
```

Figure 8.16 Creating lists.

9. An HTML Tutorial

FORMS

- Forms are used to gather data from the user.
- A form can be placed anywhere within the body of the HTML document.
- Forms may include text boxes, radio buttons, drop-down lists, and command buttons, among other objects.
- The form is usually filled out by the user and then submitted by clicking on a command button.

9. An HTML Tutorial FORMS

- Since HTML documents do not include procedural programming code, it is necessary to create a separate file to handle the data being submitted on the form.
- This separate file may include the SQL commands to log the data into the database.
- Different technologies are available for implementing this.
 - Microsoft has developed Active Server Pages (ASP) as the Windows standard technology for implementing code to handle data submitted via a Web page. The HTML page simply passes the data to the .asp file for processing.

9. An HTML Tutorial

FORMS

- The area on a form where data is entered is called a ***form element***.
- Typical elements include text boxes, password boxes, check boxes, radio buttons, and submit buttons.
- An element is created on a form object by declaring the following attributes:
 - INPUT TYPE. This attribute declares the kind of element to be placed on the form.
 - Typical TYPES include “text” (for text boxes), “password” (for password boxes), “checkbox”, “radio”, and “submit”.
 - All forms must include an input type “submit”, which is displayed as a command button on the form.
 - When a user clicks the command button, the input type “submit” transfers the input data to the ASP file that will process it.

9. An HTML Tutorial

FORMS

- NAME. This attribute assigns a variable name to the input entered into the INPUT TYPE. The .asp file identifies the input by its variable name.
- Note the following structure to the form elements:

Element Label <INPUT TYPE="element type" NAME="input variable name">

- The leading string, Element Label, that appears before the element type declaration is the label for the element.
- It will appear in the browser window as a label next to the element.
- The element is identified within the brackets following the label.
- Figure 8.18 illustrates a text box, a password box, and a submit button. Following the element type, a variable name is given. The input provided by the element will be assigned to that variable name.

9. An HTML Tutorial

FORMS

- When the data are passed from the input element to the server for processing, the value of the input will be referred to by reference to the variable name.
- A Submit button has an optional VALUE argument. The VALUE argument is the label that will appear on the button. The default VALUE argument is “Submit Query.”
- A text box has some useful attributes that can be used to control the user input. The *SIZE* attribute sets the size of the text box in terms of the number of characters that is visible.
 - In Figure 8.18, the size is 20 characters long, which is also the default size when the size argument is not used.
- The *MAXLENGTH* attribute sets the maximum length of the character string that may be entered. If the MAXLENGTH attribute is not used, it defaults to 255.

9. An HTML Tutorial FORMS

HTML Exercise 8.8:

We wish to create a form like that of Figure 8.2b.

1. Open Notepad and create the file ***UserID.html*** as shown in Figure 8.18. Save the file in your folder (MyWebPage) under Type "All files" as UserID.html. Close the file.
2. Using Netscape Navigator or Internet Explorer, browse to the file ***UserID.html*** and open it in the browser window. It should look like Figure 8.19.



```
<HTML>
<HEAD>
<TITLE>Input User ID</TITLE>
</HEAD>
<BODY>

<H1 ALIGN="CENTER"><I>University Food Incorporated</I></H1><BR>
<H2 ALIGN="CENTER">Customer Identification Page</H2><BR>

<FORM>
<P ALIGN="CENTER">
Company Name <INPUT TYPE="Text" NAME="company" SIZE=20 MAXLENGTH=30><BR>
ENTER Password <INPUT TYPE="password" NAME="passwd"><BR>
<BR>
<INPUT TYPE="submit" NAME="submit" VALUE="Submit">
</FORM>
</BODY>
</HTML>
```

Figure 8.18 Creating a simple example form.

9. An HTML Tutorial FORMS

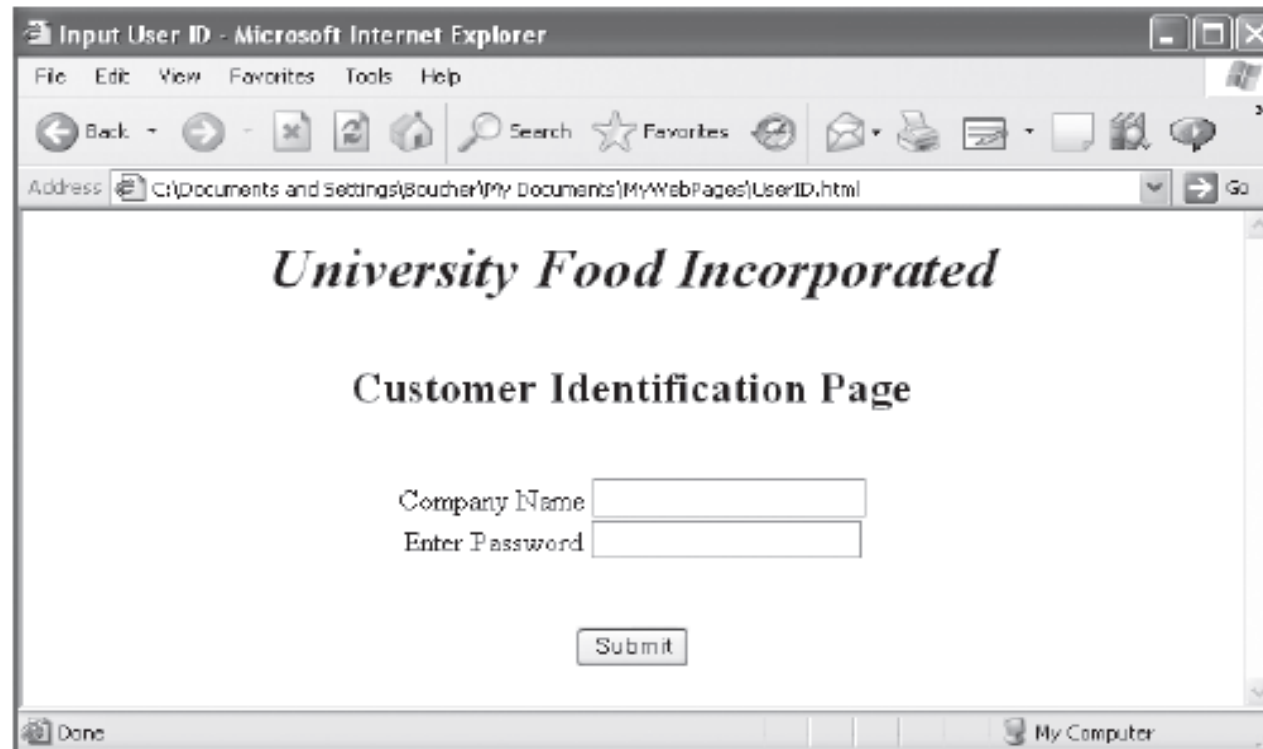


Figure 8.19 Web page created by the code shown in Figure 8.18.

9. An HTML Tutorial

FORMS

Linking a Form to an ASP File

- As previously mentioned, a form will not have any functionality unless it is linked to an ASP file.
- The reader can test this fact by opening the form UserID in a browser window (Figure 8.19), entering data, and clicking on the “Submit” button. Nothing will happen.
- Linking the form to an ASP file is done in the opening <FORM> declaration tag by adding two parameters as follows:
 - ACTION. This attribute is a pointer to the .asp file that will operate on the input data.
 - METHOD. This keyword calls a method that will be used to pass the form’s input data to the .asp file.

9. An HTML Tutorial

FORMS

Linking a Form to an ASP File

- Here we will illustrate the GET method. Using this method, when the ASP file is called, the data are appended to the URL of the .asp file identified in the action statement.
 - There is also a POST method, which is suitable for forms that send a large quantity of data to the .asp file. We will not discuss the POST method here.
 - Next, we will see the addition of these two parameters within the body tag of the form.
 - The user should open the file UserID.html and complete the code as shown in Figure 8.20.
- The .asp
- File name “response.asp” has not yet been written, so nothing will execute at this time. We have now arrived at the point where we can begin to discuss the ASP technology and the interaction between Web pages and databases.

9. An HTML Tutorial

FORMS

Linking a Form to an ASP File



```
UserID - Notepad
File Edit Format View Help
<HTML>
<HEAD>
<TITLE>Input User ID</TITLE>
</HEAD>
<BODY>

<H1 ALIGN="CENTER"><I>University Food Incorporated</I></H1><BR>
<H2 ALIGN="CENTER">Customer Identification Page</H2><BR>

<FORM ACTION="http://localhost/Example2/response.asp" METHOD="GET">
<P ALIGN="CENTER">

Company Name <INPUT TYPE="Text" NAME="company" SIZE=20 MAXLENGTH=30><BR>
Enter Password <INPUT TYPE="password" NAME="passwd"><BR>
<BR>

<INPUT TYPE="submit" NAME="submit">

</FORM>

</BODY>
</HTML>
```

Figure 8.20 Adding an action and method.